Data Mining: Social Network Mining

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Dept. of Computer Science and Engineering

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What Are Social Networks?

- A social network represents social actors and the relationships among them.
- Representing these relationships as a graph helps us analyze complex social structures.



Even a small group can have a complex web of relationships.

From Social Relationships to a Graph

- If A likes B, but B likes C who likes D and E who both like A who doesn’t even know that D exists, should F try to have G talk to B so E will know that C likes D and E, and that C will pound H if she comes around again butting in.
  - *August 1981 Peanuts comic strip by Charles M. Schulz*

Social Network Definition

- A social network can be represented as a graph:

$$G = (V, E)$$

- V: a set of vertices (nodes), representing social actors.
  - individuals, organizations, user accounts
- E: a set of edges representing relationships.
  - friendship, communication, collaboration, trust
- Edges may be:
  - Directed or undirected: following vs. mutual friendship.
  - Weighted or unweighted: communication frequency vs. the presence of a connection.

Social Network Analysis

- Social network analysis studies the structure and dynamics of social relationships.
  - It represents social actors as nodes and their relationships as edges.
- Typical questions:
  a. Which actors occupy important positions in the network?
  b. What communities exist in the network?
  c. Which new links are likely to form?
  d. How does information spread through the network?

Social Network Mining

- Social network mining discovers useful patterns and builds predictive models from network structure and attributes.

① Centrality Analysis
  - Identify structurally important nodes.

② Community Detection
  - Find groups with more internal connections than external connections.

③ Link Prediction
  - Predict missing or future edges.

④ Information Diffusion
  - Model how information spreads through a network.

Social Network Mining

① Centrality Analysis

② Community Detection

③ Link Prediction

④ Information Diffusion

Why Centrality Matters

- The Law of the Few - *Malcolm Gladwell, The Tipping Point*
  - A small number of actors can play a disproportionate role in spreading information.
- Network position can matter.
  - Some actors have many direct connections.
  - Some bridge different groups.
  - Some can reach others through short paths.
- How can we quantify these structural differences?
  - Centrality measures capture different aspects of a node’s position.

Structural importance does not necessarily imply actual influence.

Social Roles (Connectors)

- Know many people across different social circles.
- Connect individuals from different communities.
- Help information travel between groups.
- Example: A student who spreads news about a campus event across departments and clubs.

Social Roles (Mavens)

- Gather and evaluate information in a particular domain.
- Enjoy sharing useful knowledge with others.
- Help others make informed decisions.
- Example: A student who compares venues and equipment and shares practical advice with the organizers.

Social Roles (Salesmen)

- Communicate ideas persuasively.
- Build enthusiasm and address hesitation.
- Encourage others to adopt an idea or take action.
- Example: A student who builds excitement about the event and persuades others to participate.

These roles can overlap; centrality measures focus on network position.

Centrality Analysis

- Which actors occupy important positions in a social network?
  - Importance depends on what we want to measure.
- Centrality analysis assigns scores to nodes based on their network positions.
  a. Direct connections
  b. Roles in connecting other nodes
  c. Distances to other nodes
  d. Connections to other important nodes
- Different measures may rank the same nodes differently.

Different Notions of Importance

- What does "important" mean?

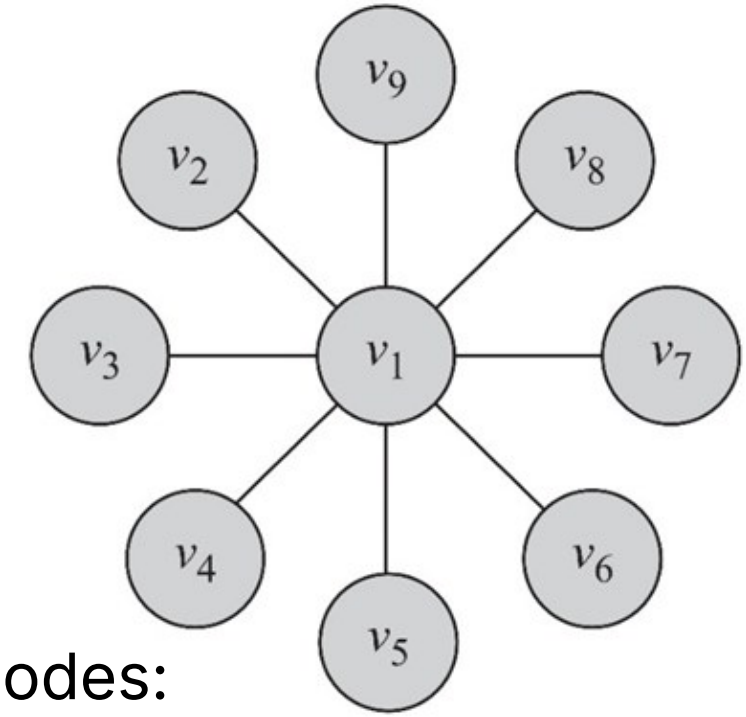
a. Who has the most direct connections?
b. Who lies on many shortest paths between other nodes?
c. Who has the smallest average distance to other nodes?
d. Who is connected to other important nodes?

Centrality Measures

a. Many direct connections → Degree Centrality
   - Number of direct connections
b. On many shortest paths → Betweenness Centrality
   - Sum, over node pairs, of the fraction of shortest paths passing through a node.
c. Small average distance→ Closeness Centrality
   - Inverse of the average shortest-path distance to other nodes
d. Connections to important nodes → Eigenvector Centrality
   - Connections to nodes that themselves have high centrality

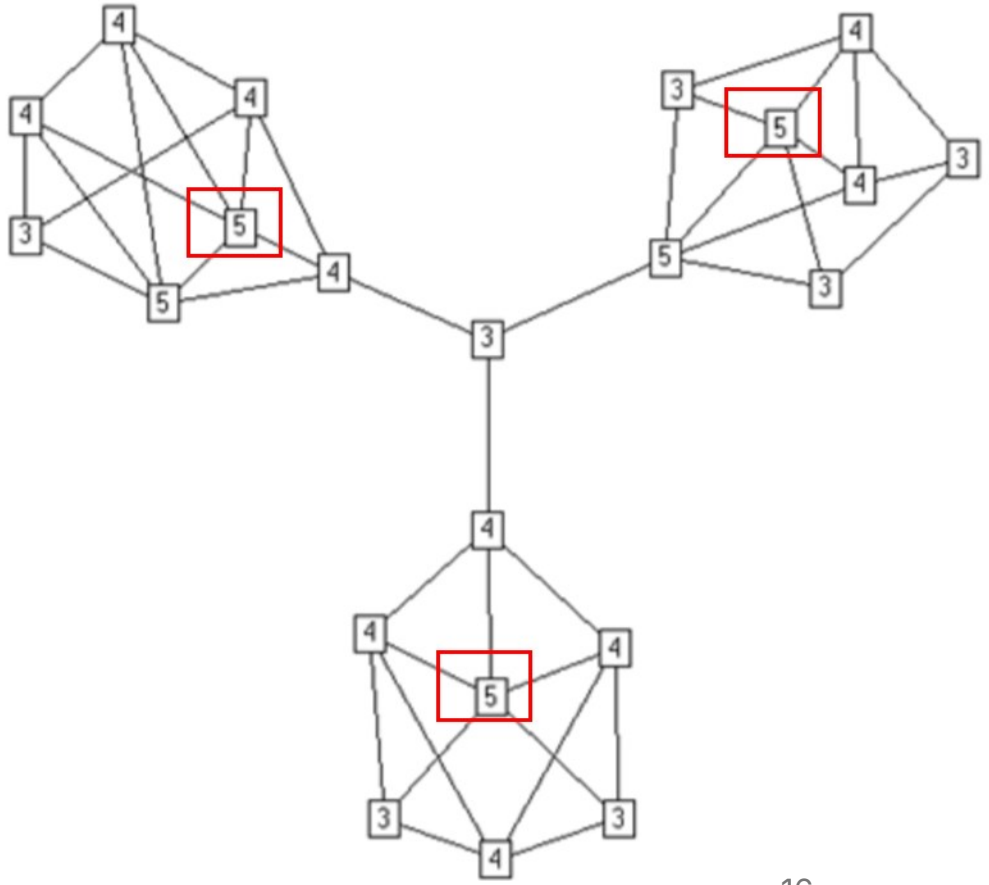
a. Degree Centrality

- Nodes with more direct connections are often considered more important.
  - More neighbors
  - More direct access to others
  - Higher potential influence
- Definition: $C_D(v) = degree(v)$
- Normalization: $C_D(v) = \frac{\deg ree(v)}{n-1}$
  - For a simple undirected graph with $n = |V| > 1$ nodes:
    - A node can connect to at most $n - 1$ other nodes.



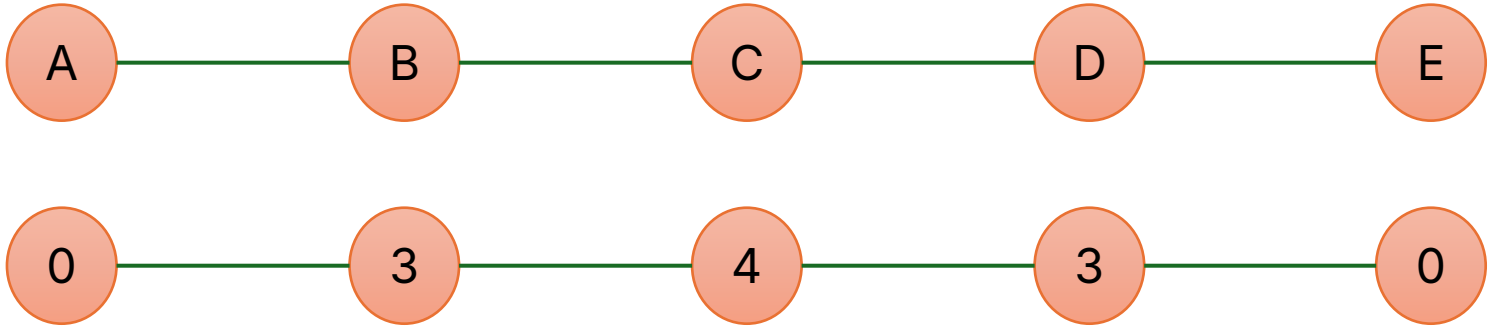
Degree Centrality (Limitations)

- Degree centrality only considers local connectivity.
- A node may have:
  - Many immediate neighbors but poor access to the rest of the network.
- High degree does not always imply global importance.
- Which nodes connect different parts of the network?
  - Maybe importance is not just about how many neighbors you have.



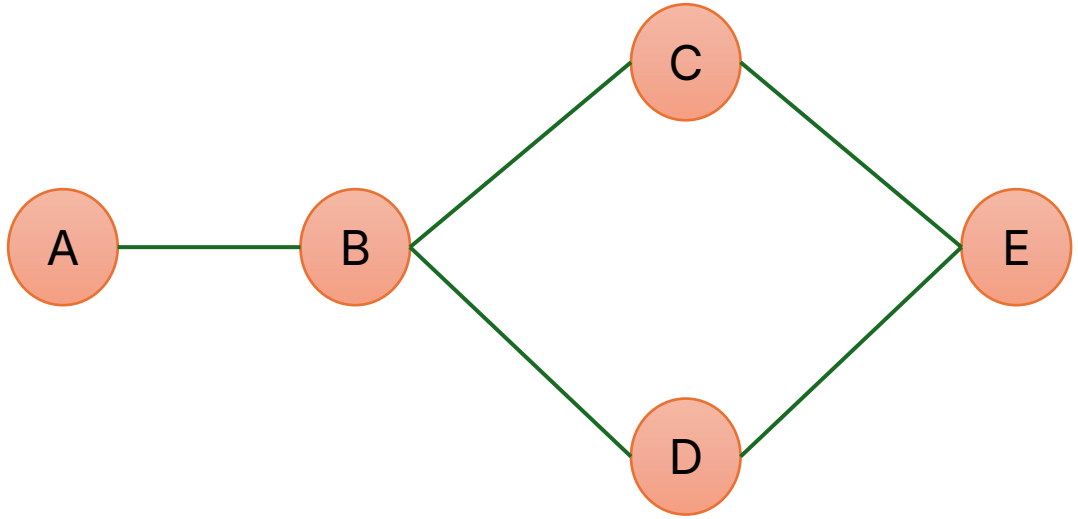
b. Betweenness Centrality

- How often does a node lie between other nodes?
  - A node has high betweenness when it lies on a large fraction of shortest paths across node pairs.
  - Acts as a bridge / broker
  - May have low degree but high strategic importance.
- Controls information flow
- Connects different parts of the network

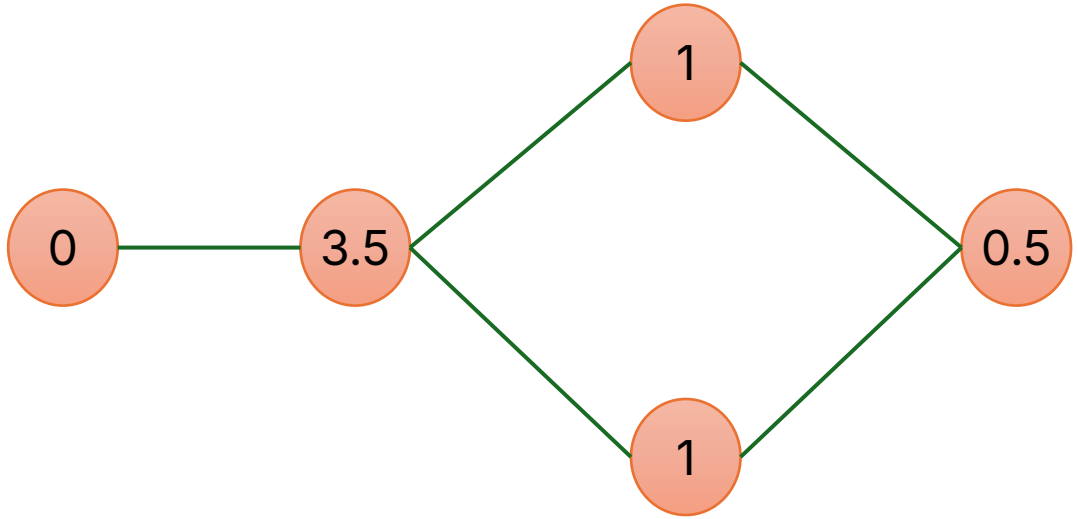


$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Betweenness Centrality (cont.)



For each node pair, credit is split equally among its shortest paths.



c. Closeness Centrality

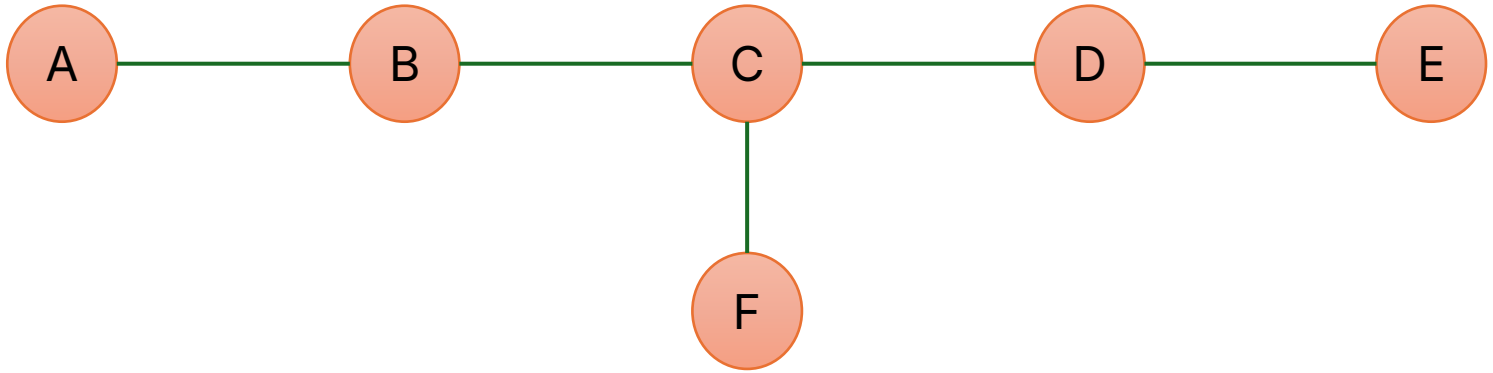
- What if importance is not about having many friends (Degree) or being a bridge (Betweenness)?
  - A node can still be important if it is close to everyone else: global accessibility.
- Characteristics:
  - Has a small average shortest-path distance to other nodes.
  - May support efficient communication when information travels along short paths.
  - Closeness depends on graph distances, not the visual position of a node.

Closeness Centrality (cont.)

- Closeness centrality measures how close a node is to all other nodes based on shortest-path distance.

$$C_C(v) = \frac{1}{\sum_{u \neq v} d(v,u)}, \text{ Normalized: } C_C(v) = \frac{n-1}{\sum_{u \neq v} d(v,u)}$$

  - $d(v,u)$ is the shortest-path distance between nodes v and u.



Vertex	C_C
A	5/13
B	5/9
C	5/7
D	5/9
E	5/13
F	5/11

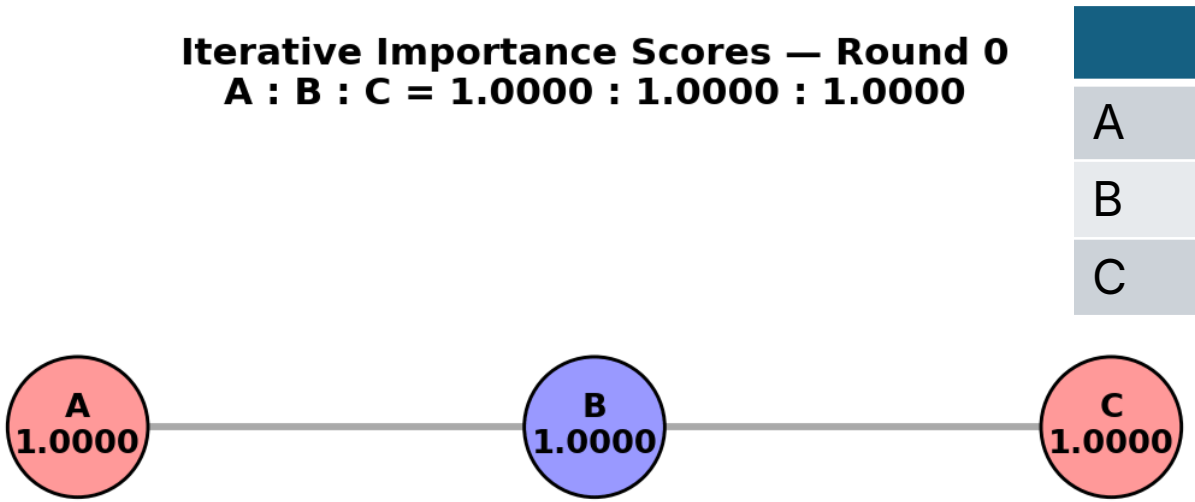
High-closeness nodes can reach the rest of the network through relatively short paths.

d. Eigenvector Centrality

- A node is important if it is connected to other important nodes.
- Comparison with degree centrality:
  - Degree centrality counts all neighbors equally.
  - Neighbors with higher centrality scores contribute more to a node’s score.
- A node with fewer neighbors may score higher if those neighbors have high centrality.
- Not just how many friends you have, but who your friends are.

How Importance Scores Stabilize

- Keep your current score and add your neighbors' scores.
- Update all nodes simultaneously.
- Rescale all scores by the same factor after each round.



	Score	Neighbor
A	0.7071	1
B	1	1.4142
C	0.7071	1

$Scale: 1.4142$

New score = own score + neighbors' scores
Then divide all scores by the largest score.

Eigenvector Centrality (cont.)

- Eigenvector centrality finds scores whose relative proportions remain unchanged when each node is rescored by the sum of its neighbors' scores.

$$x_i = \frac{1}{\lambda}\sum_j A_{ij}x_j \,, Ax = \lambda x$$

  - A: adjacency matrix
    - $A_{ij} = 1$ if nodes i and j are connected, and 0 otherwise.
  - x_i: centrality score of node i.
  - x: vector of centrality scores.
  - λ: largest eigenvalue of A.



1 陳留 | Degree centrality

直接相連的城池多

5 個鄰居：鄴、洛陽、濮陽、
小沛、許昌。

5 / 50 = 0.1000

2 長安 | Betweenness centrality

許多城池之間的最短路徑經過這裡

在西北、巴蜀與中原之間，
具有路徑中介角色。

0.2403

3 宛 | Closeness centrality

到其他城池的平均步數最少

平均距離 3.50 條城際連線；
距離以步數計算。

1 / 3.50 = 0.2857

4 許昌 | Eigenvector centrality

與高分城池相連

鄰居包含陳留、汝南、宛、
洛陽與譙；分數遞迴決定。

0.3698 (L2-normalized)

Social Network Mining

① Centrality Analysis
② Community Detection
③ Link Prediction
④ Information Diffusion

Real-World Networks Often Exhibit Community Structure

- Friendship networks
  - Family, school friends, lab members, and gaming friends.
- Citation networks
  - Papers tend to cite other papers in related research areas.
  - Examples: machine learning and biology.
- Connections are often more concentrated within groups than between groups.

Community Detection

- Community structure: groups of nodes with relatively dense internal connections and sparse connections between groups.
- Community detection: identifying these groups from the network’s connection patterns.
- Not all networks have a clear community structure.
- Communities may overlap or exist at different scales.

Community Detection vs. Clustering

Aspect	Clustering	Community Detection
Input	Objects represented by feature vectors	Nodes and edges
Main evidence	Similarity between attributes	Patterns of connections
Goal	Group objects with similar features	Find groups with dense internal connections relative to external connections
Student example	Group students by GPA and study habits	Group students by friendship or interaction patterns

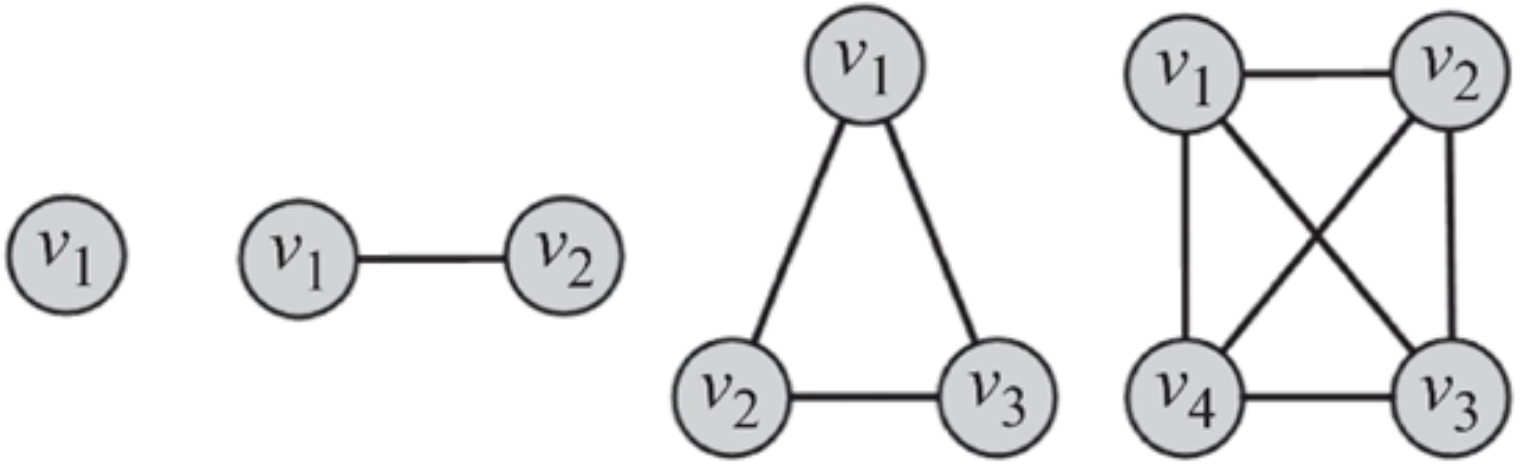
Community detection can be viewed as clustering based on network structure.

Two Approaches to Community Detection

- Member-Based: structural constraints on members
  - Cliques and distance-based variants: n-clique, n-clan, n-club
  - Degree-based constraints: k-plex, k-core
- Group-Based: group-level criteria or partition objectives
  - Graph partitioning
  - Modularity maximization

Member-Based Community Detection

- A basic structure for identifying cohesive groups is a clique.
- Every pair of distinct nodes in a clique is directly connected.
- A clique with k nodes contains $k(k-1)/2$ edges.
- Connections to nodes outside the clique are unrestricted.

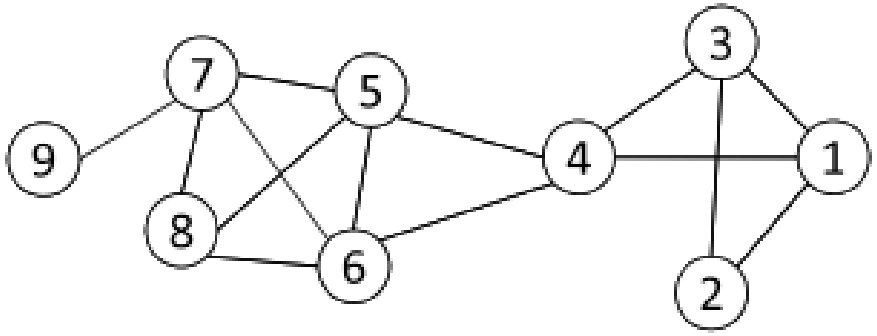


Member-Based Community Detection

- Identify candidate communities by searching for cliques.
  - Maximum clique: A clique with the largest number of nodes in the graph.
  - Maximal clique: A clique that cannot be enlarged by adding another node.
  - Every maximum clique is maximal, but not every maximal clique is maximum.
- Finding a maximum clique is NP-hard.
- The number of maximal cliques can be exponential in the number of nodes.

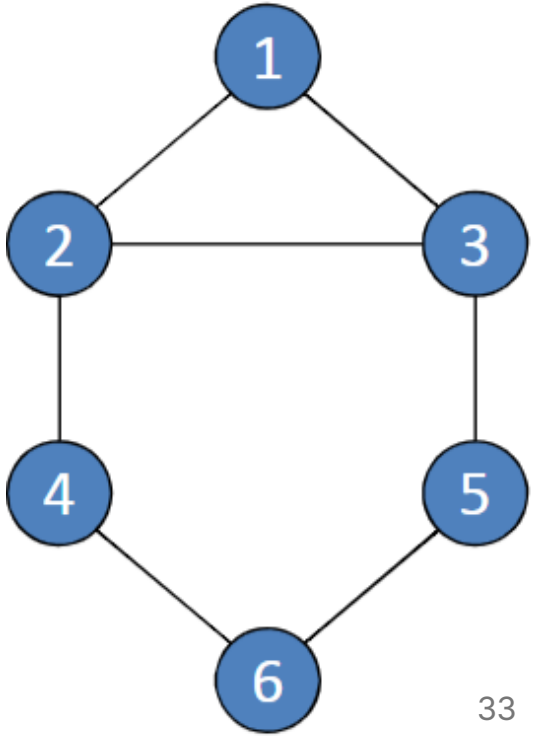
Member-Based Community Detection

- Degree Pruning: Searching for a Clique of Size $k = 4$
  - Repeatedly remove nodes whose degree in the remaining graph is less than $k - 1 = 3$.
  - ① Remove nodes 2 and 9
  - ② Remove nodes 1 and 3
  - ③ Remove node 4
  - Having degree at least $k - 1$ is necessary, but not sufficient, for membership in a clique of size k.



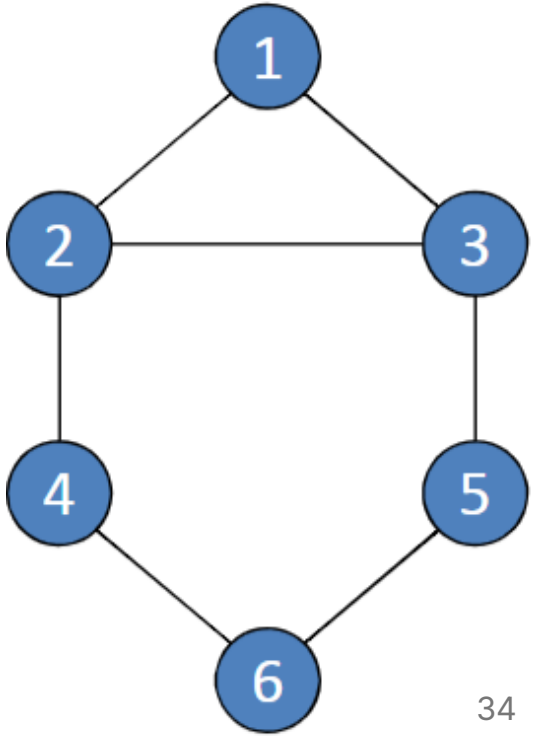
n-clique

- An n-clique is a maximal node set S such that every pair of nodes in S has distance at most n in the original graph.
  - Paths may pass through nodes outside S.
  - 1-clique: {1, 2, 3}
  - 2-cliques: {1, 2, 3, 4 ,5} and {2, 3, 4, 5, 6}



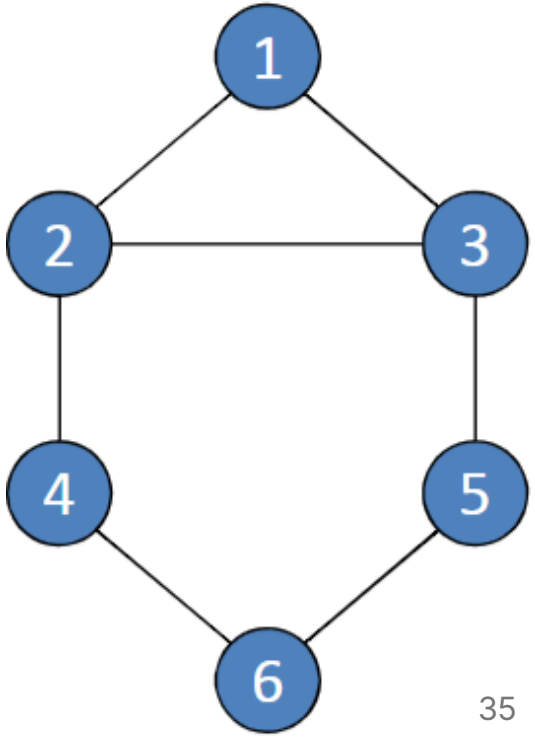
n-club

- N-club: A maximal node set whose induced subgraph has diameter at most n.
  - An n-club satisfies the distance condition of an n-clique, but may not be maximal under that condition.
- 2-clubs:
  - {1, 2, 3, 4}
  - {1, 2, 3, 5}
  - {2, 3, 4, 5, 6}



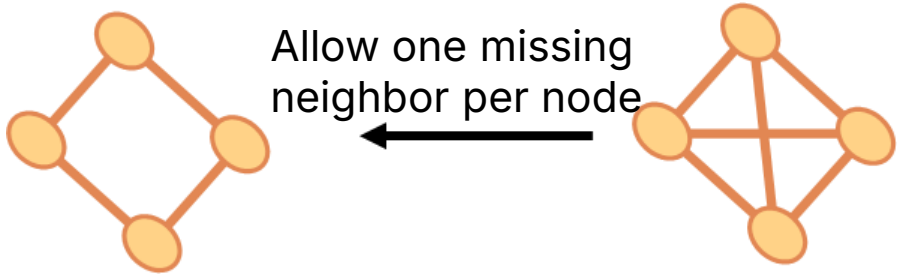
n-clan

- N-clan: An n-clique whose induced subgraph has diameter at most n.
  - 2-clan: {2, 3, 4 ,5 ,6}
  - Every n-clan is both an n-clique and an n-club.
  - {1,2,3,4,5} is a 2-clique, but not a 2-clan
    - Its internal distance between nodes 4 and 5 is 3.



K-plex and K-core

- Relaxing Cliques
  - A k-plex is a node set S in which every node has at least $|S| - k$ neighbors within S.
    - Each node may be unconnected to at most $k - 1$ other members.



  - A k-core is the maximal induced subgraph in which every node has degree at least k.

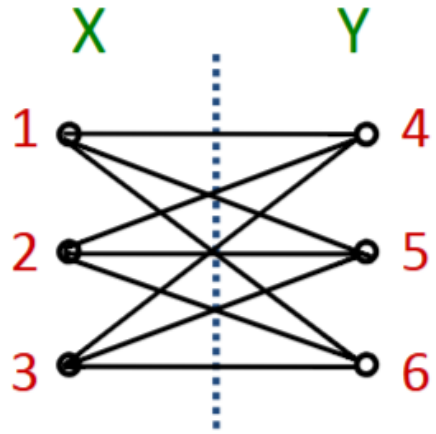
Group-Based Community Detection

- Find communities that satisfy predefined group properties.
- Evaluate groups or partitions using desired structural properties.
- Possible Group-Level Criteria
  - Balanced: Communities have similar sizes
  - Robust: Remains connected after some nodes or edges are removed.
  - Modular: Dense internally, sparse externally
  - Dense: High internal edge density
  - Hierarchical: Communities exist at multiple scales

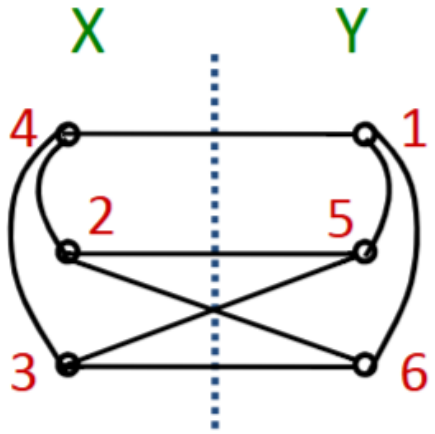
Different methods emphasize different criteria. Communities need not satisfy all of them.

Kernighan–Lin (KL) Algorithm: Balanced Graph Partitioning

- Partition a graph into two balanced groups while minimizing connections between groups.
  - KL is a heuristic that improves a balanced partition by swapping nodes between groups.
  - For a graph with $2n$ nodes, maintain two groups of n nodes each.



Original Cut Value = 9



Optimal Cut Value = 5

Girvan–Newman (GN) Algorithm: Edge-Removal

- There are usually bridging edges that connect different communities.
  - Edges connecting different communities often have high edge betweenness.
  - ① Calculate edge betweenness.
  - ② Remove an edge with the highest betweenness.
  - ③ Recalculate betweenness in the remaining graph.
  - ④ Repeat to obtain a hierarchy of partitions.
  - Select a partition from the hierarchy, for example using modularity.

Modularity: Goodness of Partitioning

- What makes a good community partition?
  - Groups of vertices with many interior edges and few exterior edges.
- Modularity measure
  - How well does a partition capture community structure?
  - Number of edges in a subgroup s is greater than expected number of edges in s (is this partition better than random chance?)

$$Modularity: \sum_{s \in S} [(\#\ edges\ in\ s) - (expected\ \#\ edges\ in\ s)]$$

  - Expected edge:

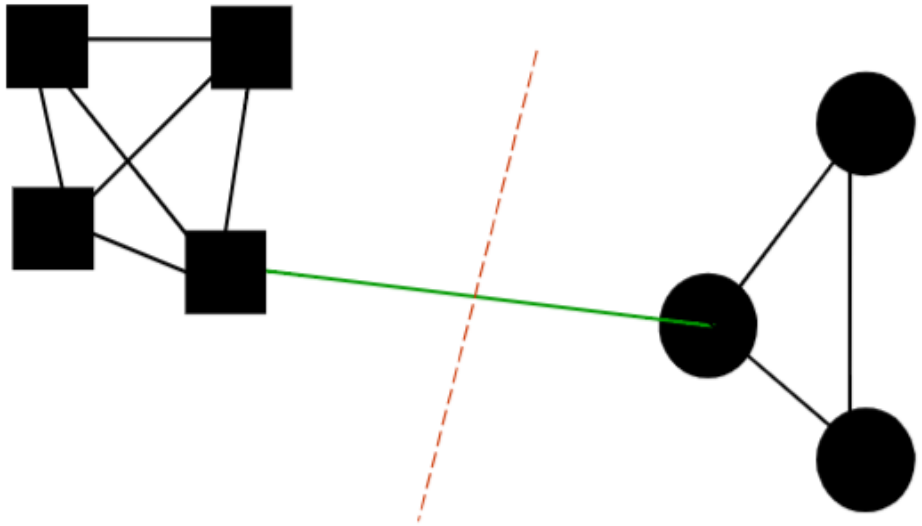
$$E(i,j) = degree(j) * \frac{degree(i)}{all\ edges} = \frac{degree(i)degree(j)}{all\ edges} = \frac{k_i k_j}{2m}$$

Modularity measure

$$Q = \frac{1}{2m}\sum_{ij}(A_{ij} - \frac{k_i k_j}{2m})\delta_{ij}$$

- m = number of edges
- A_{ij} is 1 if there is an edge between node i and j, 0 otherwise
- k_i is the degree of node i
- δ_{ij} is 1 if i and j are in same community, 0 otherwise
- Q: observed fraction of within-community edges minus the expected fraction under the null model.

Modularity: Goodness of Partitioning



$$Q = \frac{1}{2m}\sum_{ij}(A_{ij} - \frac{k_i k_j}{2m})\delta_{ij}$$

$$Q1 = \frac{1}{20}\left(12 - \frac{13*13}{20}\right) = 0.1775$$

$$Q2 = \frac{1}{20}\left(6 - \frac{7*7}{20}\right) = 0.1775$$

$$Q = Q1 + Q2 = 0.355$$

- $Q \approx 0$: close to the null-model expectation.
- $Q > 0$ → more within-community edges than expected.
- Higher Q indicates a better partition under this objective, when comparing partitions of the same graph.

Social Network Mining

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Link Prediction

- Given an observed network $G = (V, E)$, predict which unobserved links exist or which new links will form.
  a. Missing link prediction: recover hidden / unobserved connections
     - Unobserved friendships
  b. Future link prediction: predict future relationships
     - Future coauthorship links
     - Future following links
  c. Related task: link type prediction
     - Predict the type of relationship between two nodes.
     - Interaction type: collaboration vs. competition
     - Citation type: background vs. method use

Why Might Links Form?

- Homophily Principle: nodes with similar attributes tend to connect.
  - Similar interests or research topics.
- Triadic closure: nodes sharing a neighbor may form a link.
  - Two people with a mutual friend may become friends.
- These patterns motivate similarity-based link prediction.

Local Similarity-Based Link Prediction

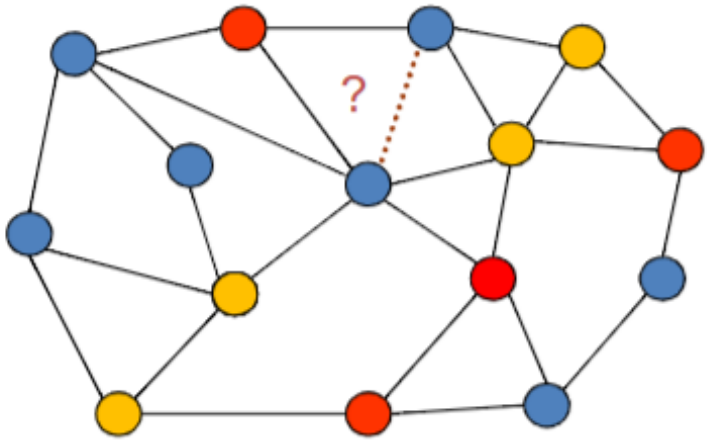
- Use local graph structure to score and rank candidate links.
  - Represent each node using its neighborhood.
  - Compute a score for each candidate node pair.
  - Rank unconnected node pairs by their scores.
- Two types of local scoring methods:
  - Neighborhood overlap: common neighbors, Jaccard similarity.
  - Degree-based scores: preferential attachment.

Common Neighbors

- Simple neighborhood-overlap score

$$Score(x, y) = |N(x) \cap N(y)|$$

  - $N(x)$: the set of neighbors of a node x
- Number of shared neighbors between x and y
  - More shared neighbors → A higher link prediction score.



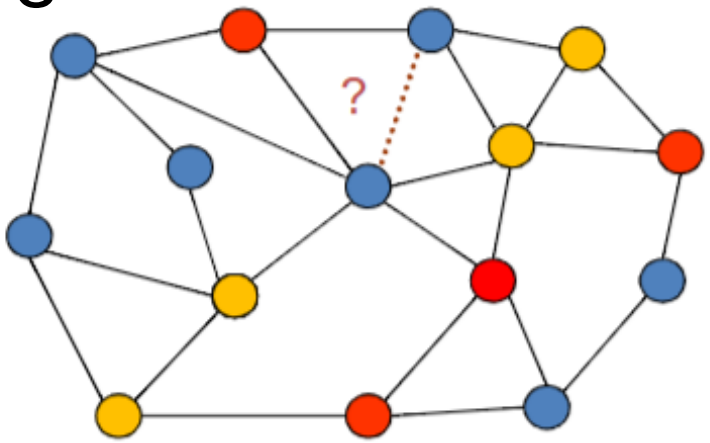
$$Score(x, y) = 2$$

Jaccard's Coefficient

- Normalize common neighbors

$$Score(x, y) = \frac{|N(x) \cap N(y)|}{|N(x) \cup N(y)|}$$

  - N(x): the set of neighbors of a node x
- Measures the fraction of shared neighbors in the union of the two neighborhoods.



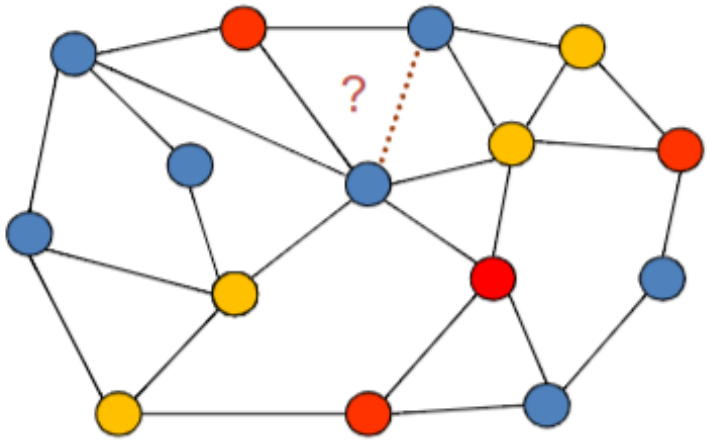
$$Score(x, y) = \frac{2}{6}$$

Adamic–Adar Index

- Give more weight to common neighbors with fewer connections.

$$Score(x, y) = \sum_{z \in N(x) \cap N(y)} \frac{1}{\log(|N(z)|)}$$

  - N(x): the set of neighbors of a node x
- Rare mutual neighbors provide stronger evidence



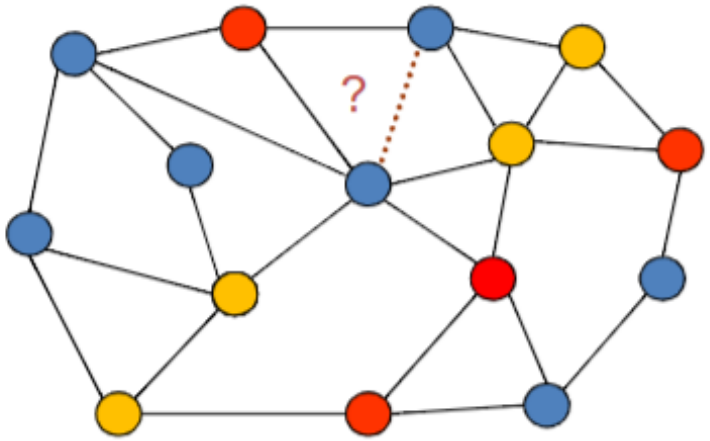
$$Score(x, y) = \frac{1}{\log 3} + \frac{1}{\log 5}$$

Preferential Attachment

- High-degree nodes attract more links

$$Score(x, y) = |N(x)| * |N(y)|$$

  - N(x): the set of neighbors of a node x
- This score favors node pairs whose endpoints have high degrees.
  - It does not require shared neighbors.



$$Score(x, y) = 3 * 5 = 15$$

Learning-Based Link Prediction

- Construct features for each candidate node pair.
  - Common neighbors, Jaccard, Adamic–Adar, and degree product.
- Train a link prediction model.
  - Positive samples: observed links.
  - Negative samples: sampled unobserved node pairs.
  - Examples: logistic regression, decision trees, or neural networks.
- Alternatively, learn node representations with a GNN.
  - Encode nodes into embeddings.
  - Use a decoder to score candidate links.

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Information Diffusion

- How do ideas, behaviors, or diseases spread through a network?
- We want to understand
  - Which nodes become activated?
  - How quickly does the process spread?
  - How large will the cascade become?
- Applications:
  - Viral marketing and product adoption
  - Rumor and news spreading
  - Epidemic modeling

Progressive Diffusion Models

- Each node is either active or inactive.
  - Initially, a seed set of nodes is active.
  - Active nodes may influence inactive neighbors.
  - Once activated, a node remains active.
- At time t_1, node v is active and node u is inactive.
  - If activation succeeds, node u becomes active at time $t_2 > t_1$.



Activation depends on the diffusion rule.

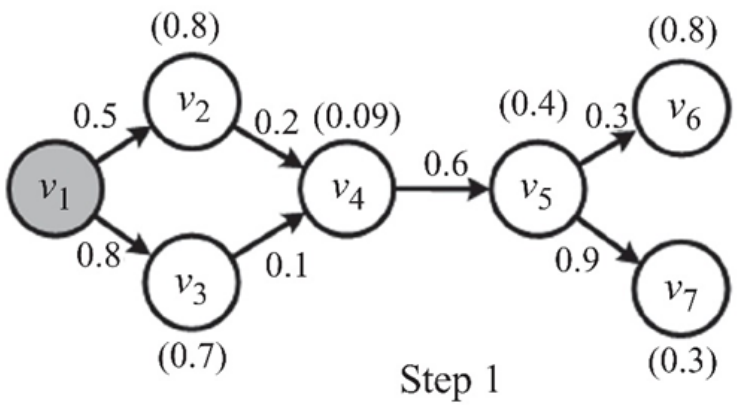
Linear Threshold Model (LTM)

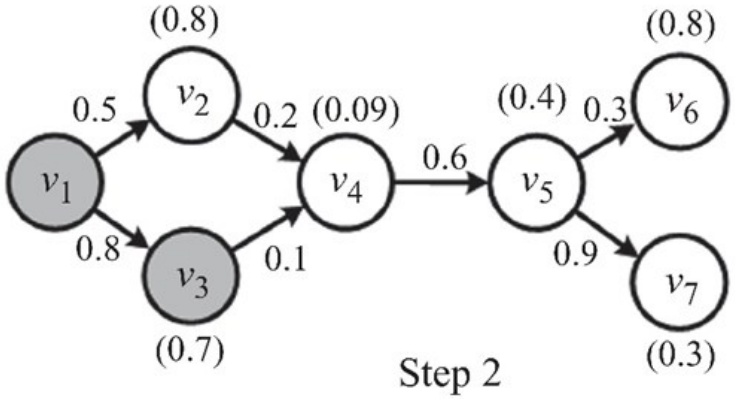
- A node becomes active when the total influence from its active neighbors reaches its threshold.
- Node *v* becomes active if:

$$\sum_{u \in N_{in}(v) \cap A_t} w_{v,u} \geq \theta_v$$

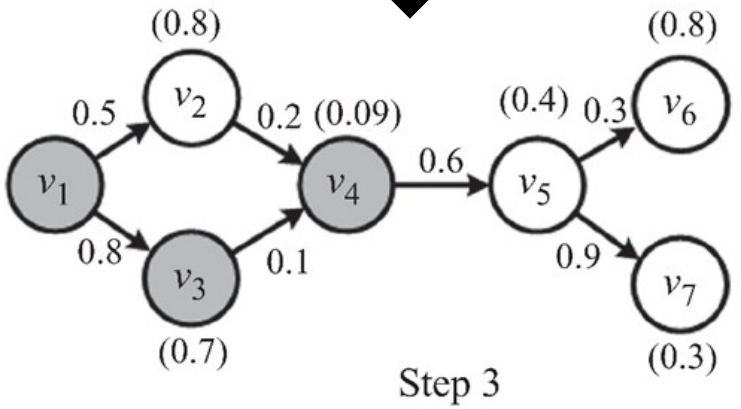
  - $w_{v,u}$: influence weight from u to v
  - θ_v : activation threshold of node v
  - Each threshold θ_v is independently sampled from Uniform $[0,1]$ once, before diffusion begins, and remains fixed.
  - All previously activated nodes remain active at each timestamp.
  - Nodes activated at step $t+1$ contribute influence starting in the next round.

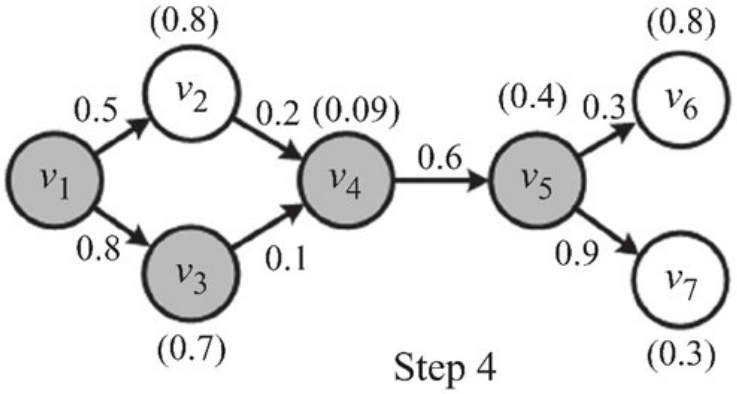
LTM Example





(Node thresholds shown in parentheses)



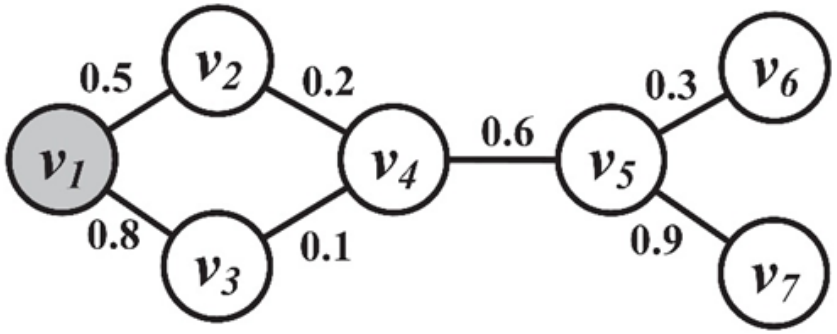


Independent Cascade Model (ICM)

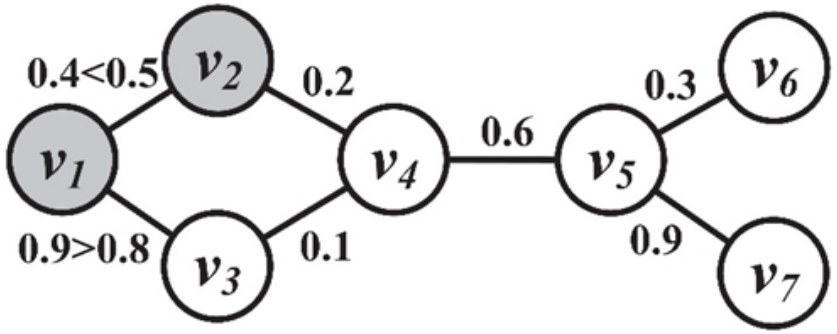
- Each newly activated node gets one independent attempt to activate each inactive out-neighbor.
  - $p(v, w)$: probability that an activation attempt from v to w succeeds.
- If node v becomes active at time t, at time step $t + 1$, it attempts to activate its neighbors w
  - Whether or not v succeeds, it cannot make any further attempts to activate w in subsequent rounds
  - Activation attempts are independent across edges.
- The process stops when no new nodes become active.

ICM Example

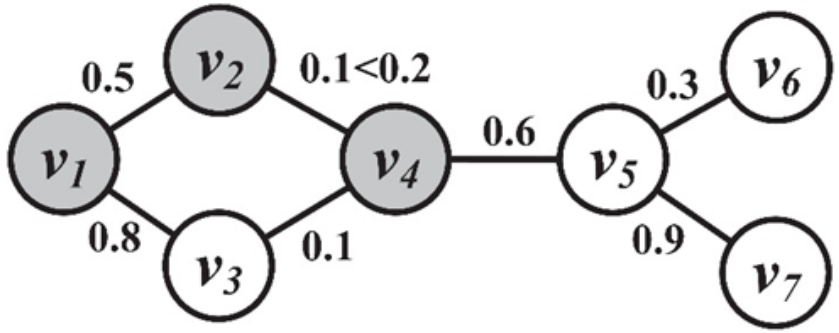
Activation succeeds if $r < p_{vw}$



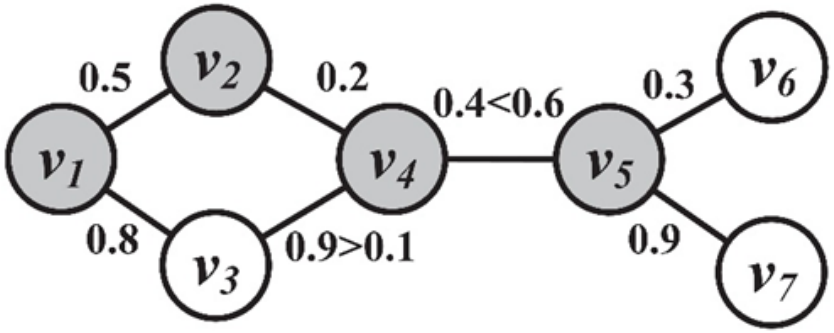
Step 1



Step 2



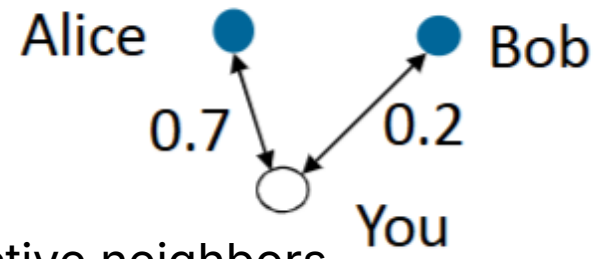
Step 3

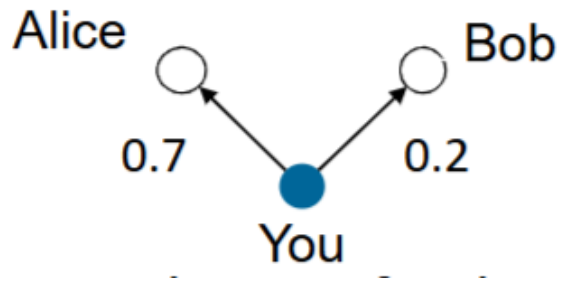


Step 4

LTM vs. ICM

	LTM	ICM
Activation rule	Total active-neighbor influence reaches a threshold	An independent activation attempt succeeds
Edge meaning	Influence weight	Activation probability
Intuition	Combined influence from peers	Individual activation attempts
Across rounds	Reevaluate using all currently active in-neighbors	One attempt per directed edge

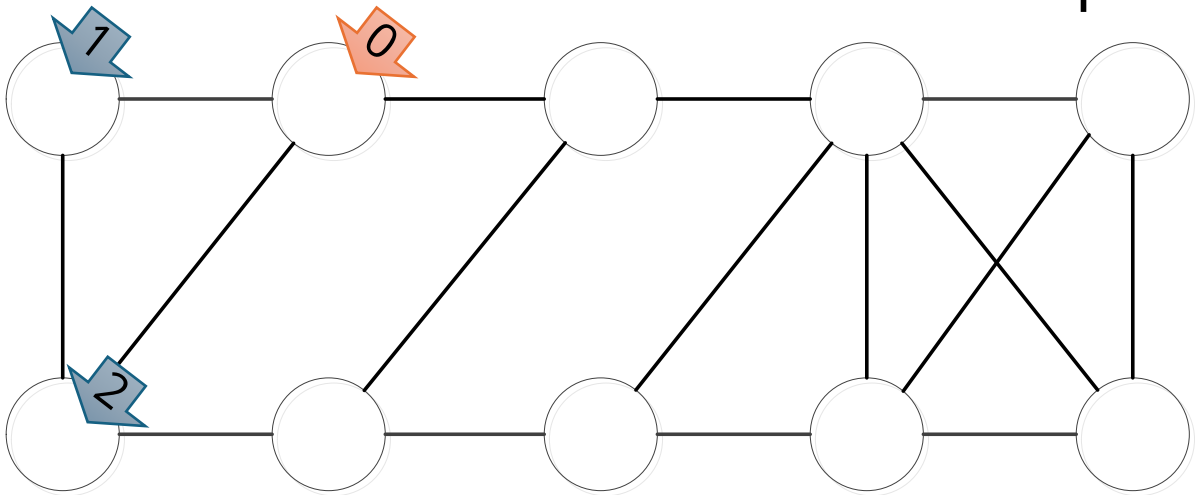


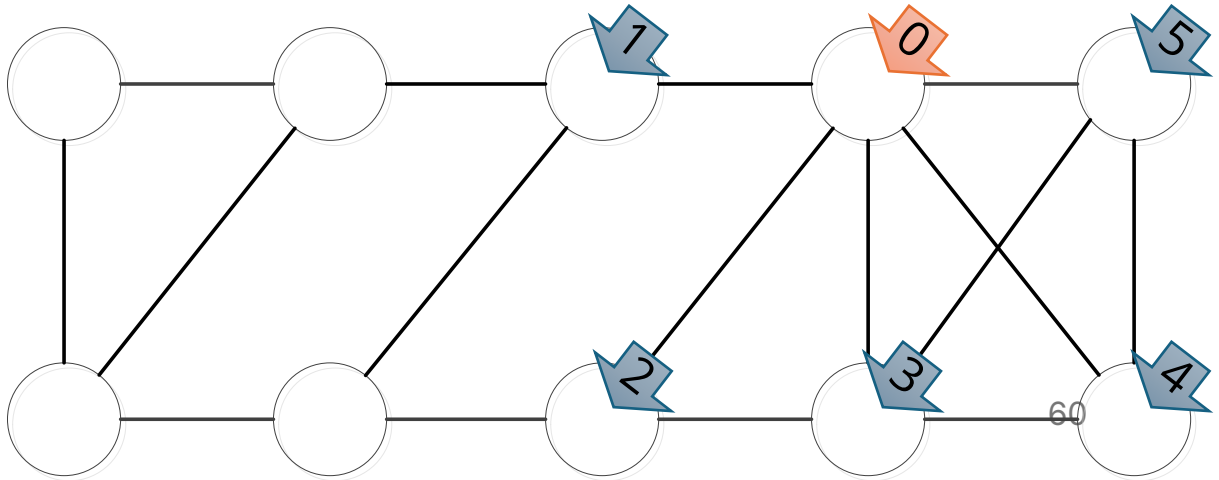


LTM combines influence from active neighbors
ICM makes independent activation attempts along edges.

Influence Maximization

- Find a small set of seed nodes that maximizes information spread
- Problem: Given a budget k, find a k-node set S to maximize $f(S)$
  - $f(S)$: An expected number of activated nodes at the end of the cascade
  - LTM/ICM: define diffusion dynamics
  - Influence Maximization: optimize seed selection under that model





Influence Maximization

Algorithm 1 Maximizing the spread of cascades – Greedy algorithm

Require: Diffusion graph $G(V, E)$, budget k

1: **return** Seed set S (set of initially activated nodes)
2: $i = 0$;
3: $S = \{\}$;
4: **while** $i \neq k$ **do**
5: $v = \arg\max_{v \in V \setminus S} f(S \cup \{v\})$;
`or equivalently` $\arg\max_{v \in V \setminus S} f(S \cup \{v\}) - f(S)$
6: $S = S \cup \{v\}$;
7: $i = i + 1$;
8: **end while**
9: Return S;

Basic Epidemic Models

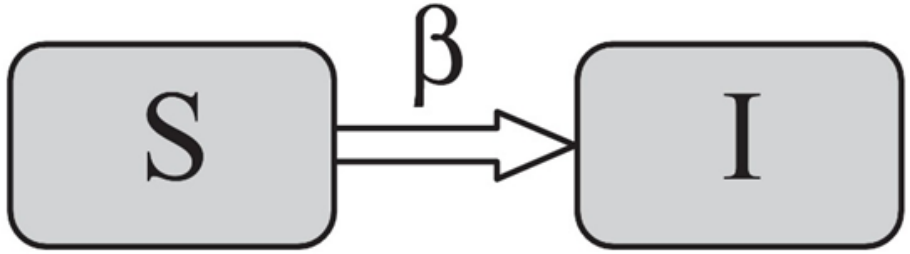
- S: Susceptible, I: Infected, R: Recovered/Removed
  a. SI Model: Infected individuals remain infected; no recovery.
  b. SIR Model: Recovered/removed individuals do not become susceptible again.
  c. SIS Model: Recovery provides no lasting immunity.
  d. SIRS Model: Immunity is temporary; individuals can become susceptible again.
  - Unlike progressive activation models, some epidemic models allow nodes to recover or become susceptible again.

a. SI Model

- Susceptible (S)
  - Uninfected individuals who can become infected.
- Infected (I)
  - Infected individuals who can infect others.
- Transition: S → I
  - Once infected, remain infected throughout the model; no recovery is included.

SI Model

- Assume a closed, homogeneously mixing population with no recovery.
  - $N = S(t) + I(t)$: constant total population
  - $S(t)$: number of susceptible individuals at time t
  - $I(t)$: number of infected individuals at time t
  - β: transmission coefficient
    - Probability that infection is transmitted during contact between a susceptible and infected individual



SI Model (cont.)

- New infections depend on:

$$\frac{dS}{dt} = -\beta SI, \qquad \frac{dI}{dt} = \beta SI$$

- Example:
  - $S = 90, I = 10,\ \beta = 0.01$: $\beta SI = 0.01 * 90 * 10 = 9$
  - At this instant, the infection rate is 9 individuals per unit time.

SIR Model

- Susceptible (S)
  - Uninfected individuals who can become infected.
- Infected (I)
  - Infected individuals who can infect others.
- Recovered/Removed (R):
  - Individuals who no longer transmit the infection and do not become susceptible again within the model.

SIR Model

- β: transmission coefficient
  - Probability that infection is transmitted during contact between a susceptible and infected individual
- γ: per-capita recovery/removal rate
  - Each infected individual leaves the infected state at rate

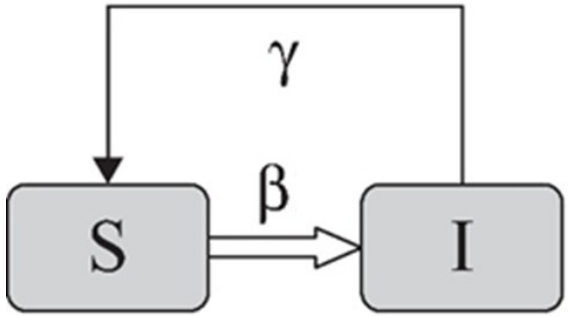
$$\frac{dS}{dt} = -\beta SI, \qquad \frac{dI}{dt} = \beta SI - \gamma I, \qquad \frac{dR}{dt} = \gamma I$$



SIS Model

- Susceptible individuals may become infected
- Infected individuals may recover
- Recovery does not provide immunity; recovered individuals immediately become susceptible again.

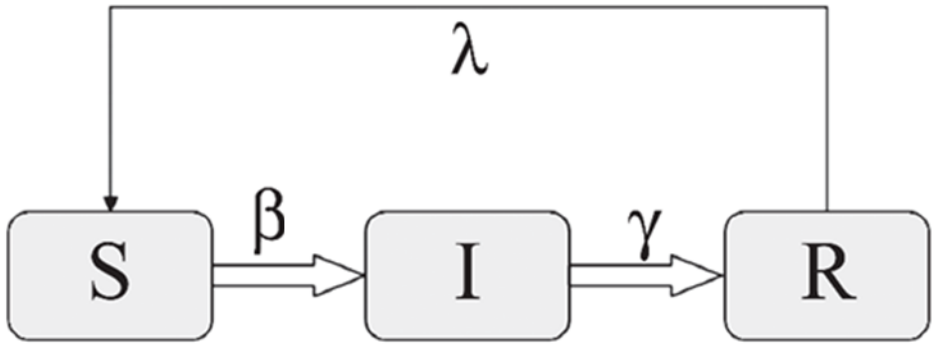
$$\frac{dS}{dt} = +\gamma I - \beta SI, \qquad \frac{dI}{dt} = \beta SI - \gamma I$$



SIRS Model

- Recovered individuals may lose immunity and become susceptible again.
  - β: transmission coefficient, γ: recovery rate, λ: immunityloss rate

$$\frac{dS}{dt} = +\lambda R - \beta SI, \qquad \frac{dI}{dt} = \beta SI - \gamma I, \qquad \frac{dR}{dt} = \gamma I - \lambda R$$



Summary

- Networks are everywhere.
  - Social media
  - Communication systems
  - Recommendation platforms
  - Biological interactions
- Network mining helps us understand the structure and dynamics of social, technological, and biological systems.

Summary (cont.)

- Social network mining studies
  - Which nodes are structurally important? Centrality Analysis
    - Rank nodes using different notions of structural importance.
  - Which groups exist in the network? Community Detection
    - Identify communities from connection patterns.
  - Which links are missing or likely to form? Link Prediction
    - Score and predict unobserved or future connections.
  - How does information spread? Information Diffusion
    - Model propagation and select seeds to maximize expected spread.